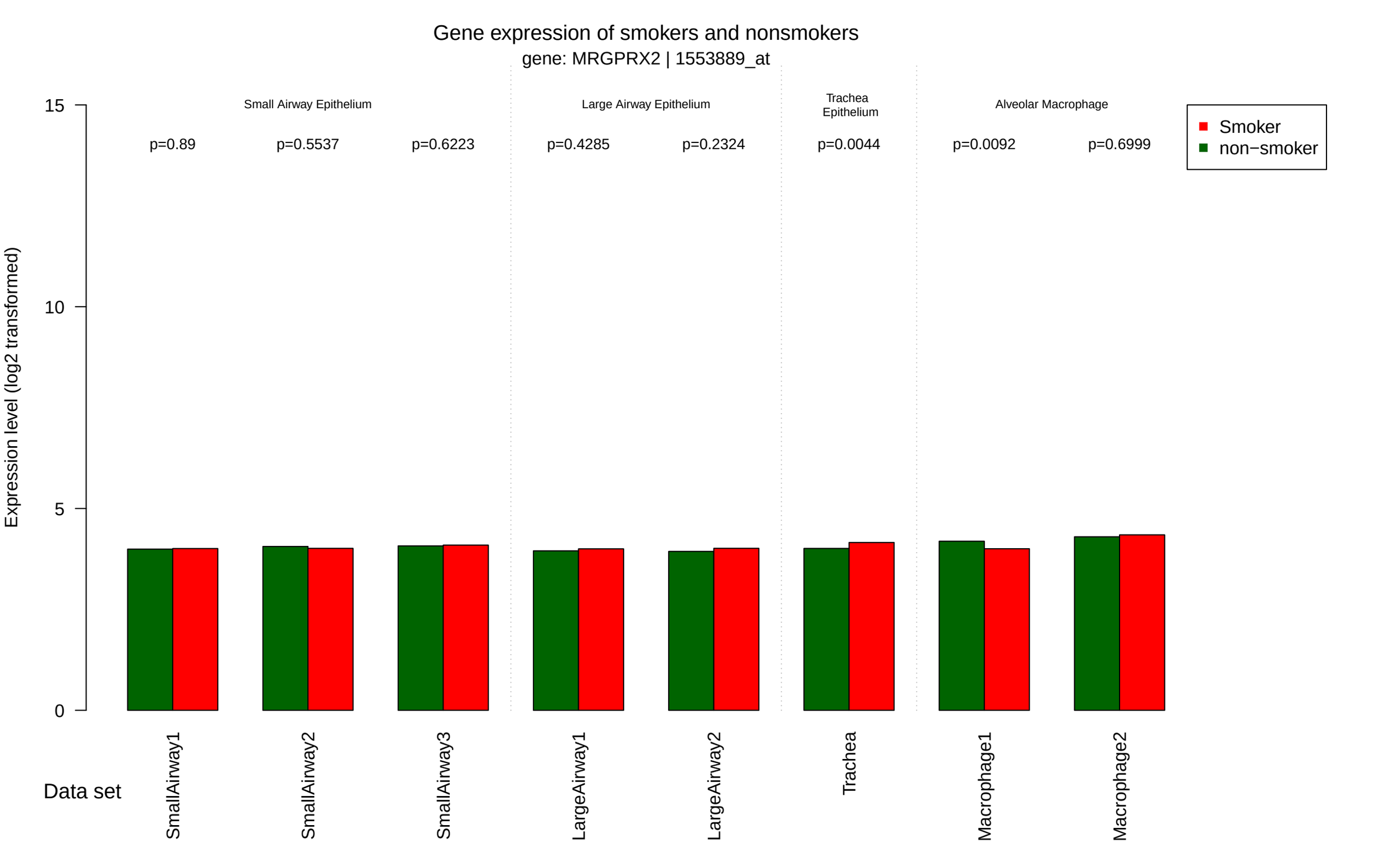
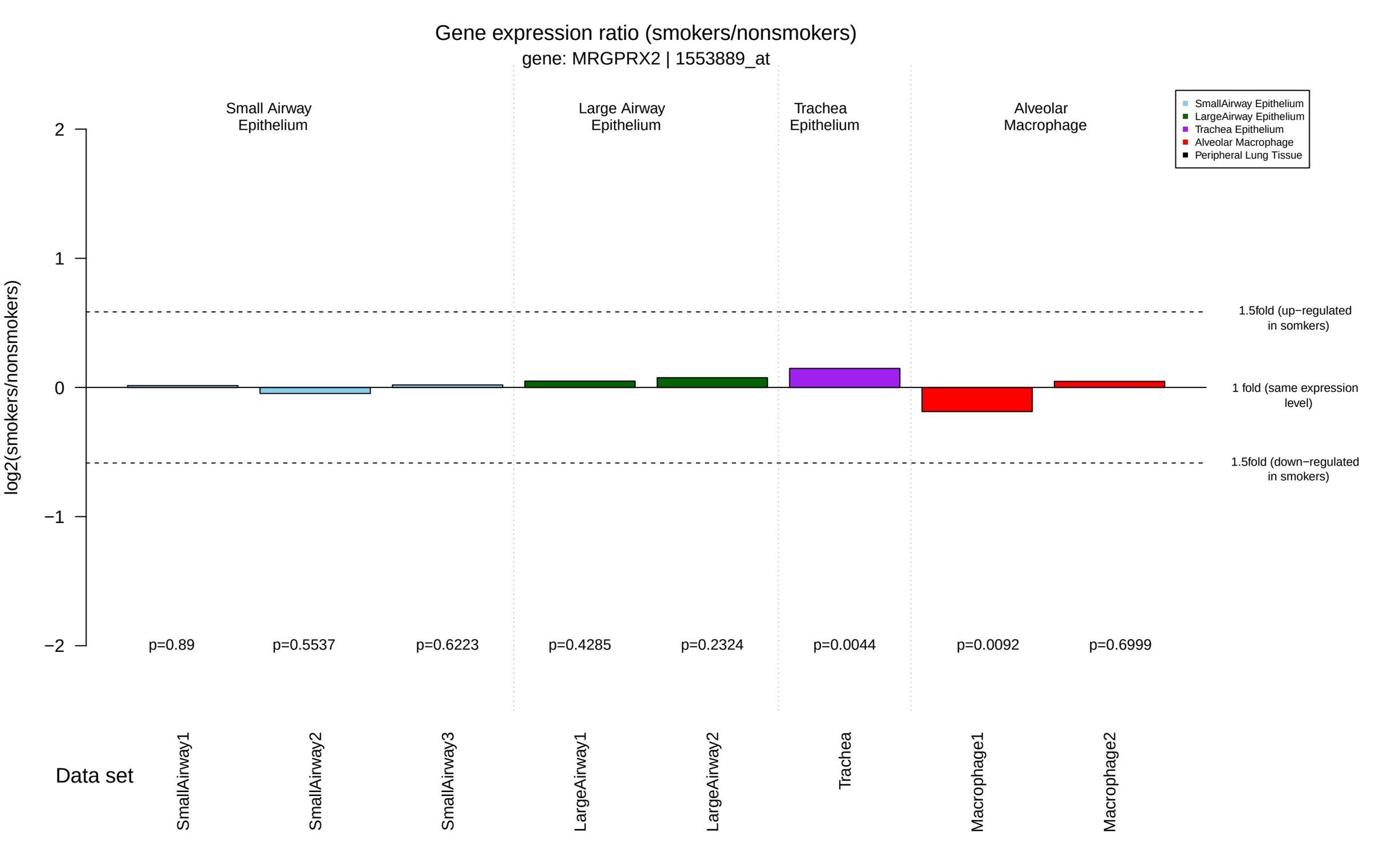
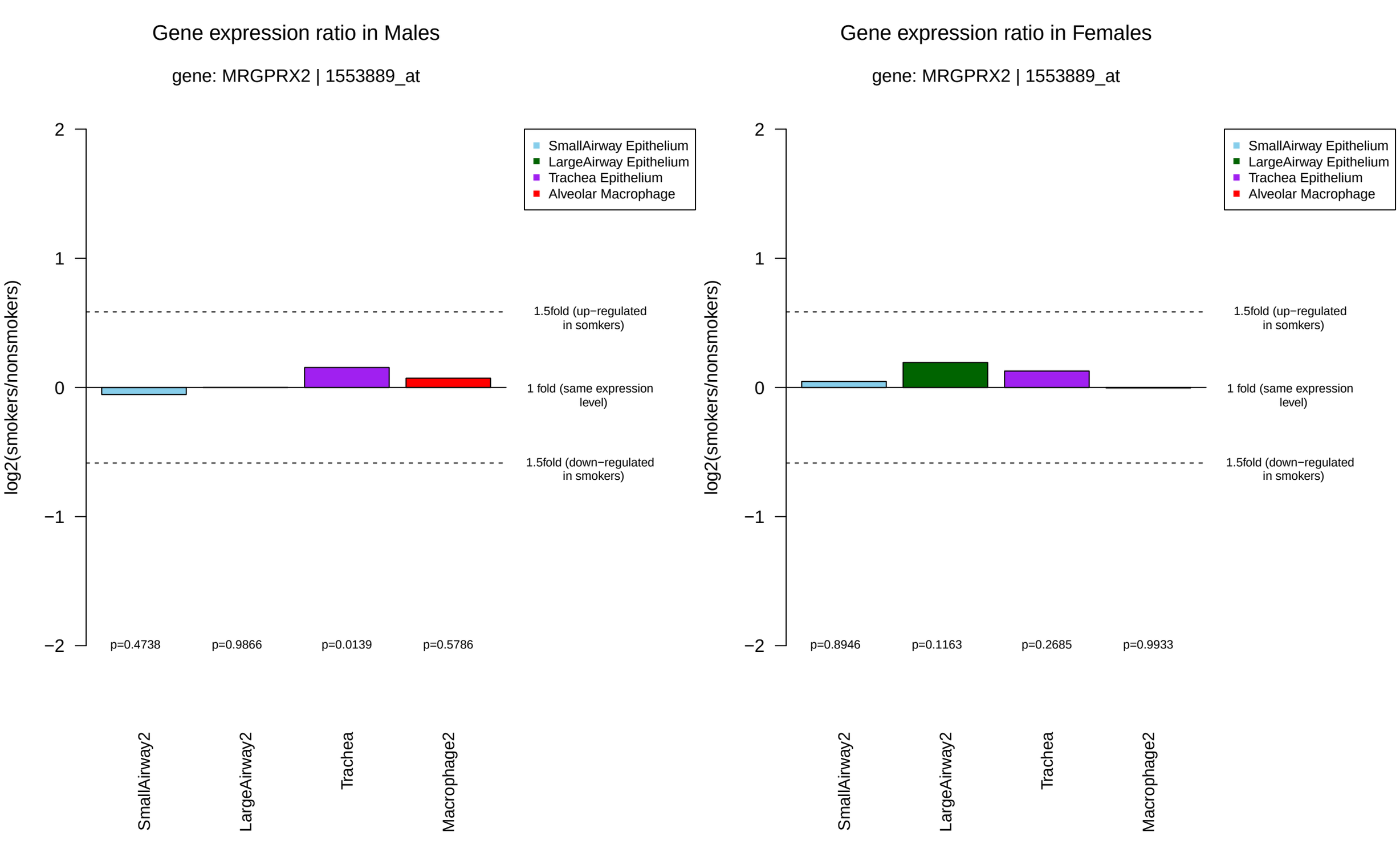
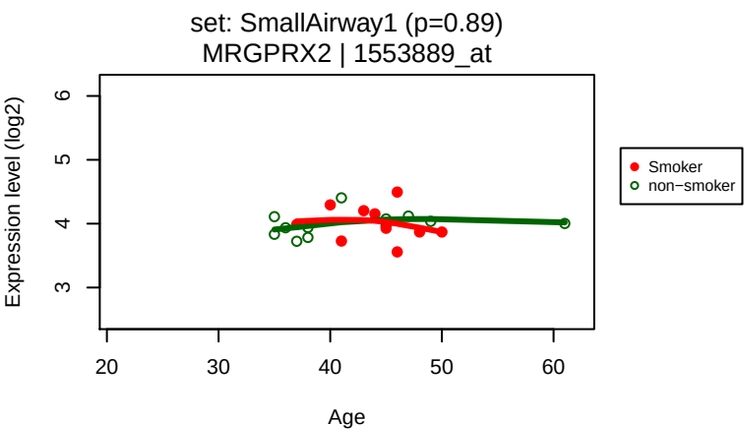
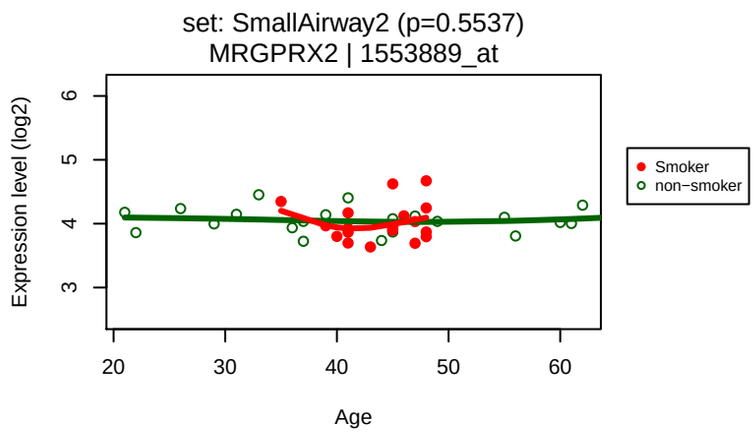
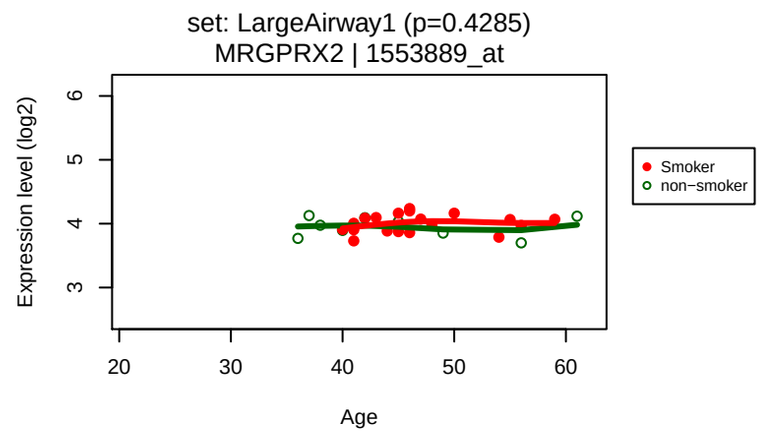

Gene expression of smokers and nonsmokers
gene: MRGPRX2 | 1553889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.89
p=0.5537
p=0.6223
p=0.4285
p=0.2324
p=0.0044
p=0.0092
p=0.6999
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

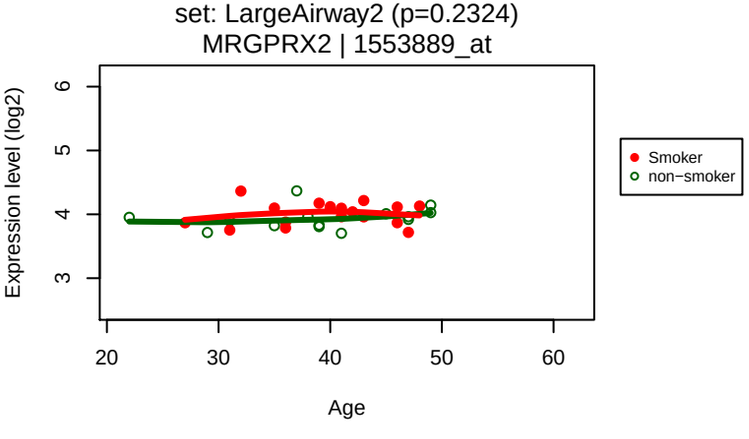

Gene expression ratio (smokers/nonsmokers)
gene: MRGPRX2 | 1553889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.89
p=0.5537
p=0.6223
p=0.4285
p=0.2324
p=0.0044
p=0.0092
p=0.6999
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

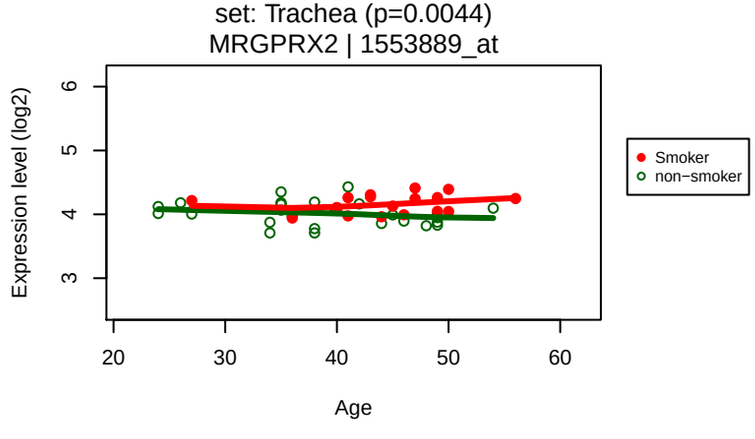

Gene expression ratio in Males
gene: MRGPRX2 | 1553889_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4738
p=0.9866
p=0.0139
p=0.5786
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRGPRX2 | 1553889_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8946
p=0.1163
p=0.2685
p=0.9933
SmallAirway2
LargeAirway2
Trachea
Macrophage2

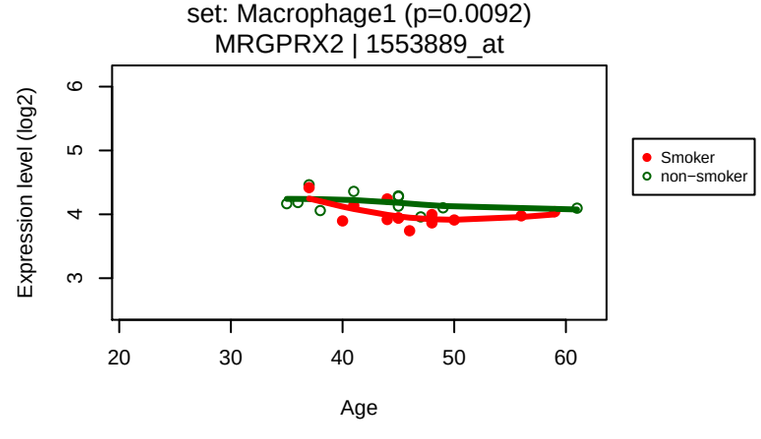

set: SmallAirway1 (p=0.89)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker

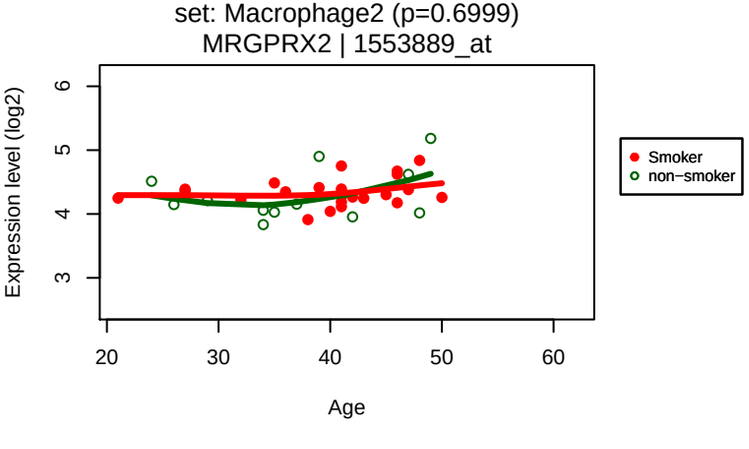

set: SmallAirway2 (p=0.5537)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker

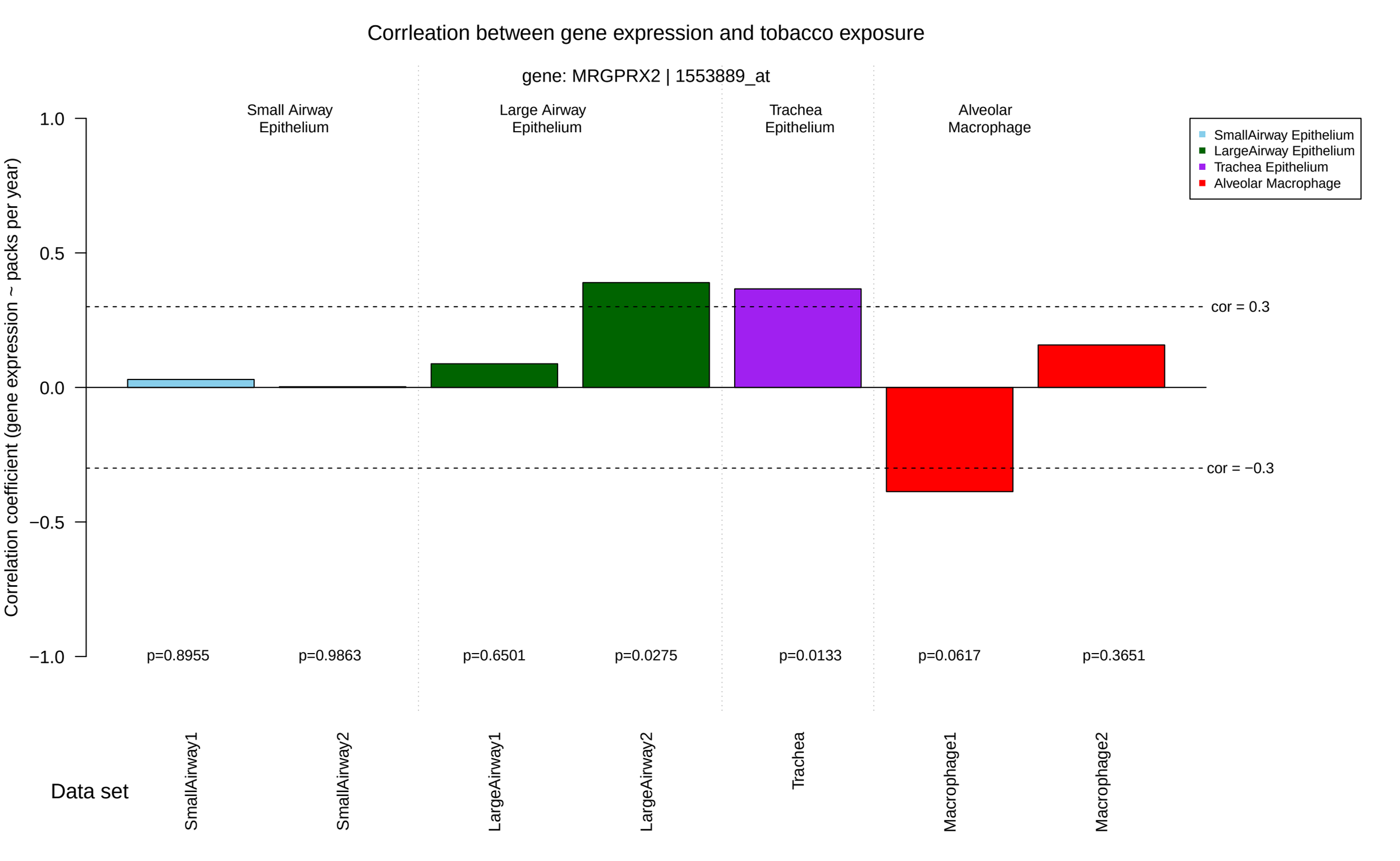

set: LargeAirway1 (p=0.4285)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2324)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0044)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0092)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6999)
MRGPRX2 | 1553889_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRGPRX2 | 1553889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8955
p=0.9863
p=0.6501
p=0.0275
p=0.0133
p=0.0617
p=0.3651
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2